MTD321

Examination - January Semester 2026

Computer Graphics and Interaction

Monday, 20 April 2026 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **FIVE (5)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is **Closed-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

Answer all questions. (Total: 100 marks)

Question 1

(a) Illustrate with a diagram to show the computer graphics system. Your diagram must include **TWO (2)** examples of each component in the system.

(8 marks)

(b) Appraise and list **FOUR (4)** differences between raster graphics and vector graphics. State examples of **ONE (1)** raster graphics file format and **ONE (1)** vector graphics file format.

(6 marks)

(c) Explain and illustrate with a diagram the texture subsampling technique. State **TWO (2)** advantages of texture subsampling over texture mipmapping.

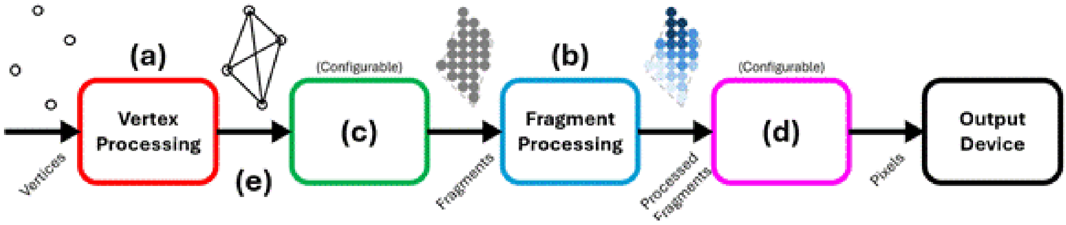
(5 marks)

(d) Describe how velocities and accelerations are used in physics-based animation to achieve (i) uniform velocity motion, (ii) acceleration motion, and (iii) deceleration motion.

(6 marks)

Question 2

(a)



Label the **FIVE (5)** parts (a, b, c, d, e) of the graphics pipeline.

(5 marks)

(b) Explain and illustrate with a diagram what is the view transformation process in the graphics pipeline and state **ONE (1)** analogy.

(8 marks)

(c) Describe and explain the order of execution of the shading and texturing processes in the graphics pipeline.

(5 marks)

(d) State the parametric representation of a straight line in 3D space. Explain why parametric representation is preferred in computer graphics over explicit and implicit representations.

(7 marks)

Question 3

(a) Explain the effects of applying the following transformation matrix.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 \\ 0 & 0 & 3.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(5 marks)

(b) Explain the effects of applying the following transformation matrix.

$$\begin{bmatrix} cos32° & 0 & sin32° & 0 \\ 0 & 1 & 0 & 0 \\ -sin32° & 0 & cos32° & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(5 marks)

(c) Explain the effects of applying the following transformation matrix.

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(5 marks)

(d) Explain and illustrate with diagrams how a complex transformation of rotating about an arbitrary point in 2D space is performed.

(7 marks)

Question 4

Analyse the codes below and answer the following questions.

```
1  boolean type = true;
2
3  void setup() {
4    size(800, 400);
5    rectMode(CENTER);
6    noFill();
7    stroke(0, 0, 0);
8    strokeWeight(3);
9  }
10
11 void draw() {
12   background(255, 255, 255);
13   if (type) {
14     int pos = height / 4;
15     for (int i = 0; i < 3; i++) {
16       rect(width / 2, pos, width, 40);
17       circle(width / 2, pos, 40);
18       pos += height / 4;
19     }
20   } else {
21     int pos = 0;
22     while (pos <= width) {
23       circle(pos, height / 2, 400);
24       pos += 400;
25     }
26   }
27 }
28
29 void mouseClicked() {
30   if (mouseButton == RIGHT) {
31     type = !type;
32   }
33 }
```

(a) Describe the functionalities of code lines 5, 7, 12, 17, and 31.

(5 marks)

(b) Assemble and illustrate with diagrams showing the two rendered output from these codes.

(20 marks)

(c) Explain how the codes allow a user to switch between the two rendered output.

(3 marks)

----- END OF PAPER -----